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Design and Validation of A Novel Planar 2R1T Remote Center-of-motion Mechanism Composing of Dual-Triangular and Straight-Line Linkages

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Benefiting from small incision and fast recovery, minimally invasive surgeries (MIS) exhibit great advantages in clinical operations. In such kind of surgeries, the remote center-of-motion (RCM) mechanisms play an important role owing to their special motion characteristics. This paper presents the design of a novel planar RCM mechanism of two rotational and one translational degrees-of-freedom (DOF). In the proposed design, the mobility of RCM mechanism is decomposed into one-DOF pure rotation and translation with a remote stationary point. The dual-triangular linkage and the Peaucellier–Lipkin straight-line linkage are introduced to achieve the remote rotation and translation, respectively. Inspired by the concept of virtual screw, a dual-helical differential-motion joint is particularly designed to generate the coaxial rotation and translation. A preliminary prototype is developed to validate the feasibility of the designed RCM mechanism. The experimental results show that the developed prototype is easy to control and of acceptable positioning accuracy, which manifests potential application in MIS.

Keywords: Minimally invasive surgery, RCM mechanism, Peaucellier–Lipkin linkage, dual-triangular linkage.

1 Introduction

Benefiting from the tremendous advantages over the traditional surgical procedures, robot-assisted MIS have attracted attention [1, 2], in a wide range of applications, such as endoscopy-assisted neurosurgery [3], intraocular operations [4] and MRI-guided prostate intervention [5]. In these surgical systems, the RCM mechanisms play the key role in mechanically constraining the end-effector to a stationary point far from the fixed base. Hence, a variety of mechanisms with RCM characteristics have been proposed.

The most widely used RCM mechanism is the da Vinci surgical system [6], which is an One-DOF RCM linkage with double-parallelogram architecture proposed by Hamlin et al. [7]. Based on this linkage, Tayor et al. [8] further developed and proposed a laparascopic surgical robot. Because of the geometric constraint of the parallelograms, this mechanism can generate rotation around a straight point distal from the fixed base. In addition, a variety of related RCM mechanisms has been proposed based on this linkage. Zong et al. [9] proposed a systematic approach for the type synthesis of One-DOF RCM mechanisms and synthesized several concrete examples through their method. Furthermore, a variety of planar Two-DOF RCM mechanisms which are able to generate pitch and translation were proposed by Gijbels et al. [10] and Nisar et al. [11]. Based on the Two-DOF spherical linkage, Zemiti et al. [12] presented a compact and lightweight RCM mechanism. Clavel et al. [13] proposed a RCM mechanism which can generate two rotational DOF (roll and pitch) by arranging two parallelogram-based limbs in an orthogonal array. Inspired by the Peaucellier-Lipkin straight-line mechanism, Chen et al. [14] and Li et al. [15] proposed the planar Two-DOF RCM mechanisms by docking two limbs in parallel. Different from these parallelogram-based architectures, Chen and coworkers [16, 17] developed a novel One-DOF RCM mechanism which significantly reduces the space requirements.

Moreover, researchers have also focused on the alternative solutions for the synthesis of RCM mechanisms and a variety of spatial RCM manipulators were proposed. Gosselin et al. [18] proposed a 3-RRR spherical parallel mechanism to realize the RCM characteristic. Kuo and Dai et al. [19] proposed a novel Four-DOF parallel RCM mechanism with fully-decoupled end-effector motion. Wang et al. [20] proposed a family of spatial fully parallel RCM mechanism by intersecting motion planes. RAVEN robot proposed

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by Tanev [21] is a Four-DOF spatial mechanism with RCM characteristic which is guaranteed by a serial spherical limb and actuated by two extra UPS limbs. Based on the ingenious structure of the 'Agile Eye', Gosselin et al. [22] proposed a class of four-legged parallel mechanisms with 3R1T RCM characteristic. Chen et al. [23] developed a novel spatial parallel RCM mechanism which can generate four DOF by applying two limbs decoupled with each other. Recently, Li and coworkers [24] presented a comprehensive study on the type synthesis of SP-equivalent motion generators.

Despite inherent advantages of the existing RCM mechanisms, most of them have less than four DOF. Hence, extra actuators are needed on the end-effector to achieve the required 3R1T mobility [25]. As a consequence, the structure of these RCM mechanisms becomes complex and the inertia of moving bodies increases significantly. Although a few of fully parallel four-DOF RCM mechanisms have been proposed with all actuators mounted on the fixed base, the parallel structure makes them space-consuming.

On the basis of these consideration, this paper presents the design of a novel planar 2R1T (rolling, spinning, and insertion) RCM mechanism with a compact structure. In the proposed design, the required mobility of the studied RCM mechanism is decomposed into individual ones, and realized via simple planar multi-loop one-DOF linkages, respectively. Particularly, the widely-used dual-triangular linkage [16, 17] and the well-known Peaucellier–Lipkin straight-line linkage [26] are employed to achieve pure rotational and translational motions. Besides, the concept of virtual screw [27] is introduced to generate the coaxial spinning and insertion for the end-effector through the differential motion of a dual-helical joint. The main advantages of the proposed design reflect on the simplicity in structure and the easiness of realization. In addition, closed-form solutions to both the forward and the inverse kinematics problems can be derived readily.

To validate the effectiveness of the proposed design, a prototype of the proposed RCM mechanism is developed using easy-to-access materials and simple fabrication methods. And a variety of preliminary experiments, including the workspace characterization and the accuracy assessment, are conducted. The results exhibit that the developed prototype fulfills the RCM feature of motion and performs acceptable positioning accuracy. Moreover, it is easy to extend the current design to a spatial 3R1T RCM mechanism by adding a directly-actuated revolute joint mounted on the fixed base. Therefore, the proposed design has the potential application in robot-assisted MIS operations.

This paper is organized as follows: Section 2 presents the conceptual design of the proposed planar 2R1T RCM mechanism. Then, in Sec. 3, analytical solutions to both the forward and the inverse kinematics, as well as the Jacobian matrix, of the proposed RCM mechanism are derived. Based on the closed-form models of kinematics, Section 4 exhibits the characterization of the mechanism's reachable workspace and singular configurations. To validate the feasibility of the proposed design, a prototype of the 2R1T RCM mechanism is developed in Sec. 5, on which preliminary experiments are conducted. Finally, conclusions are drawn in Sec. 6.

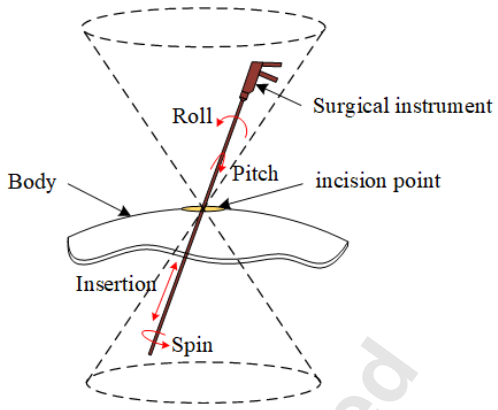


Fig. 1. Mobility requirement of general MIS operations.

2 Mechanism Design of the Planar RCM Mechanism

In general, a RCM mechanism in robotic MIS systems requires four DOFs mobility to implement the operation. As shown in Fig. 1, it can be characterized as the 3R1T 4-DOF motion, including three concurrent rotations (Roll, Pitch, and Spin), and the translational axial insertion. In generic cases, the rotation of pitch can be realized via a directly-actuated revolute joint mounted on the fixed-frame. Therefore, in this paper, only the 2R1T mobility, namely rolling, spinning and insertion, is taken into account in the design of the RCM mechanism. These motions are decomposed into simpler individual ones, which are achieved by well-known planar one-DOF closed-loop linkages, including the dual-triangular RCM linkage, the Peaucellier–Lipkin straight-line linkage, and the dual-helical differential motion joint.

2.1 Conceptual Design

As shown in Fig. 2, in the proposed design, the required mobility of RCM mechanisms is decomposed into three subchains, which are implemented by a dual-triangular (DT) linkage, two coincident straight-line (SL) mechanisms, and a coaxial dual-helical joint, respectively.

Figure 2 illustrates the main structure of the conceptual design. First, a DT linkage is mounted on the fixed frame to generate the RCM rotation (Rolling) about the z-axis. Then, two coincident SL linkages are connected to the distal end of the DT linkage to produce coaxial translations along a line through the RCM point. Meanwhile, a dual-helical joint is introduced to transform the translations of the SL linkages into coaxial rotation (spinning) and translation (insertion), through the differential motion mechanism.

With this structure, the translational motions of the SL linkages are independent from the DT linkage. As a result, in the designed RCM mechanism, the rolling of end-effector is decoupled from the spinning and insertion, which could ease the kinematic control of the mechanism. In addition, the axial spinning and insertion is achieved by the differential motion of the dual-helical joint with an invariant transmission. And this can further simplify the control theme. The details on the kinematics and implementation of these linkages will be given in the following sections.

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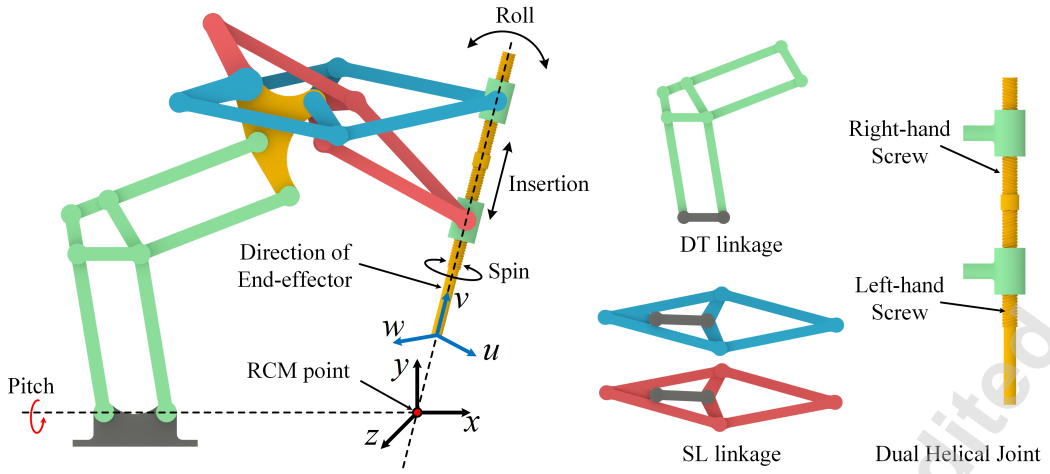


Fig. 2. Conceptual design of the proposed 2R1T RCM mechanism.

2.2 Dual-Triangular Linkage

As drawn in Fig. 3, the DT linkage [17] is a multi-loop eight-bar linkage with a symmetrical structure. It relates to a planar One-DOF mechanism, as $f = 3\times(8-1)-2\times10 = 1$. According to the geometric constraint, the configurations of this mechanism keep symmetric about the line ED. Furthermore, $\triangle OAD$ needs to be the similar triangle of $\triangle OBE$ all the time.

According to similar triangles $\triangle OAD$ and $\triangle OBE$, it is known that, the line DE will always pass through O in light of the similarity between the triangle $\triangle OAD$ and $\triangle OBE$. Moreover, according to the symmetric constraint, the distal link, more specifically the line HG, will also pass through the RCM point all the time, as shown in the figure. Since all the links of the DT-linkage is far away from the stationary point O, it has been widely used to generate the rolling of the end-effector in the RCM mechanisms [16, 17].

In the proposed design, we extend the DT linkage by replacing the distal link HG with an intermediate platform IKG, to which additional employed linkages can be attached. The line IK serves as the fixed frame of the SL linkages, so that the corresponding axial translations will be along a line perpendicular to it, as shown in Fig. 3. As a result, the shape of the link IKG can be arbitrarily adjusted as long as the line of translations generated by the SL-linkages passes through the stationary point O. Hence, the straight line OT will purely rotate around the RCM point, when the DT linkage moves. Here, the angle $\angle IOT$, and the lengths l_{IO} and l_{IK} are the key parameters confining the connecting SL-linkages, which will be discussed in the following section.

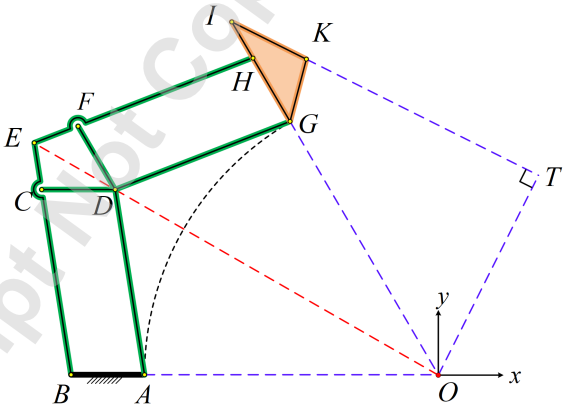


Fig. 3. Schematic of the dual-triangular linkage.

2.3 Dual-helical Joint Actuated by Dual SL-linkage

To actuate the coaxial rotation and translation for the end-effector, the concept of cylindrical drive (C-drive) [27] is employed in the proposed RCM mechanism. As shown in Fig. 4, it relates to a differential motion mechanism that transforms two collinear translations into the virtual screw motion (coaxial rotation and translation) about the same axis. The coaxial dual helical joint, as illustrated in the figure, is a simple and compact way to realize this function.

Ideally, the pitches of the helical joints in the C-drive could be opposite to each other, namely of same magnitude but with left-hand and right-handed directions, respectively. Then, a pure translation can be achieved when the two nuts move in the same direction at the same rate. While pure rotations will be generated, when the nuts move in the opposite direction at the same rate. Actually, the C-drive works in any situation as long as the pitches are different from each other. To simplify the implantation of the proposed RCM mechanism, in the prototype developed in this paper, one of the pitches in the dual-helical joint is set to zero. In such a case, it degenerates to a revolute joint which is very convenient to implement using ball bearings.

To generate the differential motion of the dual helical joint, the remarkable Peaucellier–Lipkin mechanism [26] for perfect straight-line motion is introduced to adjust the axial positions of nuts. This linkage corresponds to a planar multi-loop eight-bar mechanism with one-DOF mobility as well. Specifically, as shown in Fig. 5, the output position P will

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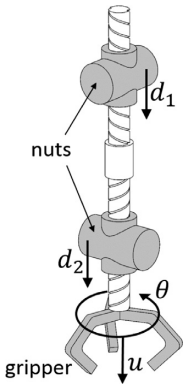


Fig. 4. Schematic of the Two-DOF active C-drive [27].

generate a perfect line that perpendicular to the base link IK, when the crank IR rotates. To achieve the characteristics of the straight-line motion, the lengths of links in this eight-bar linkage should satisfy the following condition as

$$\begin{cases} l_{IR} = l_{IK}, \quad l_{KM} = l_{KN} \\ l_{PM} = l_{PN} = l_{RM} = l_{RN} \end{cases} \tag{1}$$

By a proper design of the SL-linkage, the base link IK can be directly attached to the intermediate platform on the DT linkage, and meanwhile, the straight line PO simply coincides with the virtual one OT. As a result, the axial position of the nut can be actively controlled by the SL linkage. Here, two identical SL linkages, as shown in Fig. 2, are adopted to actuate the translation of the two nuts in the dual helical joint, respectively. Thus, the coaxial spinning and insertion of the end-effector can be actuated by the coincident SL linkages, independently from the rolling movement.

Another benefit of using the straight-line linkages is the high performance of force transmission from the cranks of the SL linkages to the differential motion of the dual helical joint. Since the linear outputs of the SL linkages are coincident with the axis of the dual helical joint, theoretically the transmission angle perfectly keeps as good as 90°, when the designed RCM mechanism moves.

3 Kinematics Modeling and Analysis

This section presents the kinematics modeling and analysis of the RCM mechanism proposed in Sec. 2. Owing

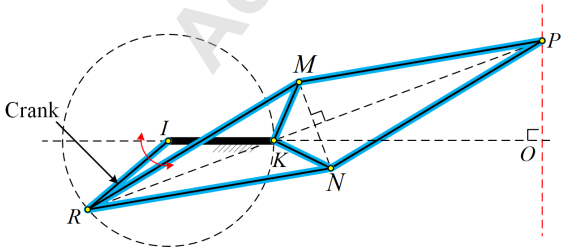


Fig. 5. Schematic of the Peaucellier–Lipkin straight-line linkage.

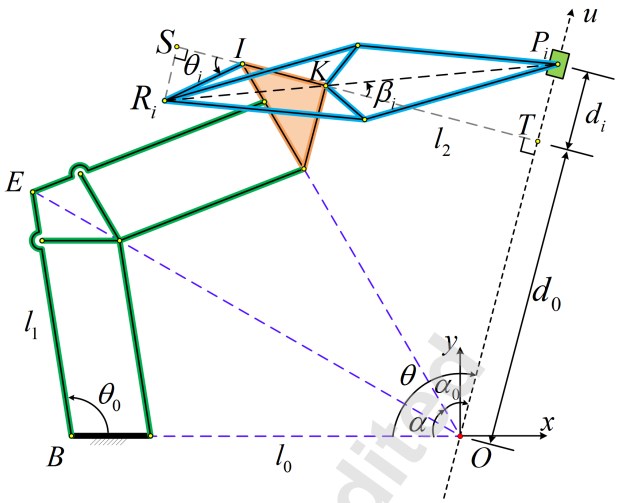


Fig. 6. Kinematics of the proposed RCM mechanism.

to the decomposition of the 2R1T RCM characteristics into three simpler linkages, closed-form models can be developed readily, for both the forward and inverse kinematics problems. As well, the Jacobian matrix is derived analytically, by means of relating the input/output relations of the composed linkages in a consecutive manner.

As addressed in the above section, the proposed Three-DOF RCM mechanism is actuated by one dual-triangular linkage and two straight-line linkages separately. Thus, the drives of these One-DOF linkages, namely θ_0, θ_1, and θ_2 as indicated in Fig. 6, are defined as the input variables of the studied mechanism. On the other hand, the rolling and spinning angles, as well as the insertion displacement, of the mechanism's end-effector, termed θ, ϕ, and l, respectively, are known as the output variables.

3.1 Forward Kinematics Model

In the forward kinematics problems, the active inputs of the mechanism's actuators are given, while the corresponding outputs of the end-effector need to be specified. For the 2R1T RCM mechanism proposed in this paper, this process can be implemented in a consecutive way, by means of analyzing the dual-triangular and straight-line linkages, and the dual-helical joint, individually.

According to motion characteristics of the DT linkage, the end-link generates a pure rotation about the RCM point. Thus, its orientation, depicted by the azimuth angle α, can be uniquely determined in terms of the corresponding input variable θ_0 as

$$\alpha = 2\arctan\frac{l_1 \sin\theta_0}{l_0 - l_1\cos\theta_0} \tag{2}$$

which is simply twice of angle $\angle BOE$, as shown in Fig. 6. Here, l_0 and l_1 correspond to the lengths of the link OB and BE, respectively.

As for the SL linkages, the motion of their end-links are strictly constrained to the line OT, which is perpendicular to

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IK. Here, T is the foot point of the line IK to OT, which is stationary fixed with respect to the end-link ΔGIK of the DT linkage. Accordingly, the outputs of these linkages, namely the positions P_1 and P_2 on OT, can be determined readily in terms of the inputs θ_1 and θ_2, respectively.

As pointed out in Sec. 2.3, in the adopted straight-line linkage, the line P_iR_i always passes through the fixed point K according to its geometric constraint. Therefore, the triangle $\triangle KTP_i$ is similar to $\triangle KSR_i$, as shown in Fig. 6. Accordingly, the output position can be obtained directly as

$$d_i = l_2 \tan\beta_i = \frac{l_2 \sin\theta_i}{1+\cos\theta_i} = l_2 \tan\frac{\theta_i}{2}, \quad i = 1,2 \tag{3}$$

where d_i is defined as the distance from T to P_i. l_2 relates to the constant distance from K to T. And β_i corresponds to the angle $\angle TKP_i$, also $\angle SKR_i$. Since the length of IR_i is same as IK, β_i is a function exclusively of the input θ_i.

Further, the spinning angle and insertion displacement of the mechanism's end-effector can be achieved by the differential motion of the dual-helical joint. As indicated in Sec. 2.3, the two coaxial helical joints are connected together and actuated via the outputs of the two PL linkages, respectively. Then, we have

$$\begin{bmatrix} \phi \\ d \end{bmatrix} = \frac{1}{p_2 - p_1} \begin{bmatrix} -1 & 1 \\ p_2 & -p_1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \mathbf{M}_p \boldsymbol{d} \tag{4}$$

where d is the distance from the center of the dual-helical joint, termed P_0 to T. ϕ corresponds to the rotation angle about the screw axis. While p_1 and p_2 are the pitches of the corresponding helical joints, respectively. For convenience of representation, $\boldsymbol{d} = [d_1, d_2]^T$ and the constant matrix $\mathbf{M}_p$ can be regarded as the transmission ratio of the differential motion generated by the dual-helical joint.

Therefore, the output variables of the whole 2R1T RCM mechanism can be derived as

$$\begin{bmatrix} \theta \\ \phi \\ l \end{bmatrix} = \begin{bmatrix} \alpha_0 + \alpha \\ \phi \\ d_0 + d \end{bmatrix} \tag{5}$$

where α_0 is the constant angle $\angle IOT$ and d_0 is the constant distance between O and T.

From Eqs. (2)∼(4), it is known that the forward kinematics of the studied RCM mechanism can be derived in a straightforward way, according to the closed-form models of the DT and straight-line linkages. Further, the rolling angle θ of the end-effector can be uniquely determined by the DT linkage. Meanwhile, the spinning angle ϕ and the insertion displacement l are functions exclusively of the two PL linkages. In other words, the 2R1T mobility of the proposed RCM mechanism is partially decoupled, which can simplify the control theme of kinematics.

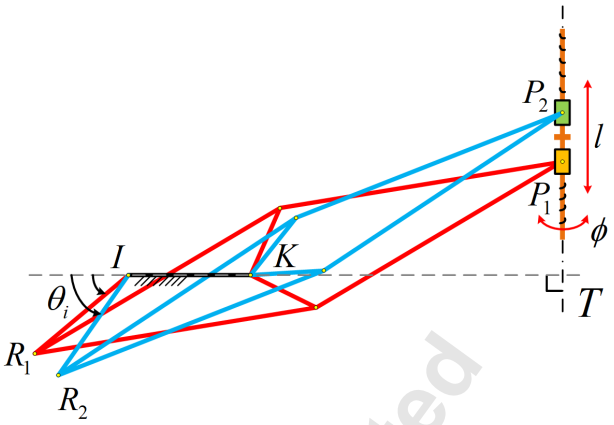


Fig. 7. Realization of the virtual screw generator (C-driver).

3.2 Inverse Kinematics Model

In the inverse kinematics problems, the output variables of the end-effector are specified for prescribed tasks. On the contrary, the corresponding input variables are required to be identified in this case. Similar to the forward problems, the inverse kinematics model of the studied RCM mechanism can also be derived in a consecutive way.

On one hand, the input angle of the DT linkage can be obtained directly by solving the trigonometric function (2) for θ_0, which yields

$$\theta_0 = \arcsin\frac{l_0 \sin\frac{1}{2}\alpha}{l_1} - \frac{1}{2}\alpha \tag{6}$$

where $\alpha = \theta - \alpha_0$ according to Eq. (5).

Analogously, the collar positions of the two straight-line linkages can also be obtained in a similar way, by solving the linear Eq. (4) directly, which yields

$$\boldsymbol{d} = \mathbf{M}_p^{-1} \begin{bmatrix} \phi \\ d \end{bmatrix} \Rightarrow \mathbf{M}_p^{-1} = \begin{bmatrix} p_1 & 1 \\ p_2 & 1 \end{bmatrix} \tag{7}$$

where $d = l - d_0$ as given in Eq. (5).

Then, the required inputs of the straight-line linkages' cranks can be determined directly by means of solving Eq. (3) for θ_i as

$$\theta_i = 2\arctan\frac{d_i}{l_2}, \quad i = 1,2 \tag{8}$$

From Eqs. (6)∼(8), it is known that the inverse model of kinematics of the designed 2R1T RCM mechanism can be achieved conveniently as the inverse kinematics of the composing One-DOF linkages is easy to derive. This is because of the utilization of the dual-triangular and the straight-line linkages, in conjugation with the differential motion via the coaxial dual-helical joint, which are particularly suitable for the requirement of RCM characteristics.

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3.3 Jacobian matrix

As presented in Sec. 3.1 and 3.2, the kinematics model (both forward and inverse) of the proposed 2R1T RCM mechanism can be constructed readily in a consecutive way. As well, the system Jacobian matrix can be derived in a similar manner. Let $\boldsymbol{q} = [\theta_0, \theta_1, \theta_2]^T$ be the vector of the mechanism's input variables, and $\boldsymbol{x} = [\theta, \phi, l]^T$ be the output one. Then, the system Jacobian matrix relating the input velocity of the actuators to the output one of the end-effector can be represented in a generic form as

$$\dot{\boldsymbol{x}} = \mathbf{J}\dot{\boldsymbol{q}} \Rightarrow \mathbf{J} = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{q}} = \begin{bmatrix} \frac{\partial\theta}{\partial\theta_0} & \frac{\partial\theta}{\partial\theta_1} & \frac{\partial\theta}{\partial\theta_2} \\ \frac{\partial\phi}{\partial\theta_0} & \frac{\partial\phi}{\partial\theta_1} & \frac{\partial\phi}{\partial\theta_2} \\ \frac{\partial l}{\partial\theta_0} & \frac{\partial l}{\partial\theta_1} & \frac{\partial l}{\partial\theta_2} \end{bmatrix} \tag{9}$$

where $\frac{\partial\theta}{\partial\theta_1} = \frac{\partial\theta}{\partial\theta_2} = \frac{\partial\phi}{\partial\theta_0} = \frac{\partial l}{\partial\theta_0} = 0$ since θ can be determined exclusively in terms of θ_0, while ϕ and l are independent of θ_0. The other non-zero entries can be derived readily according to the kinematics model.

From Eq. (9), it is known that the system Jacobian $\mathbf{J}$ degenerates to a block diagonal matrix, composing of a 1×1 block on the upper-left and a 2×2 block on the lower-right. First, the upper-left block can be derived directly by means of differentiating Eq. (2) with respect to θ_0 as

$$\frac{\partial\theta}{\partial\theta_0} = \frac{\partial\alpha}{\partial\theta_0} = \frac{2l_1\cos(\theta_0+\frac{1}{2}\alpha)}{l_0\cos\frac{1}{2}\alpha - l_1\cos(\theta_0+\frac{1}{2}\alpha)} \tag{10}$$

where α can be obtained as in Eq. (2).

According to the kinematics of the dual-helical joint (as presented in Eq. (4)), the 2×2 lower-right block can then be obtained as

$$\begin{bmatrix} \frac{\partial\phi}{\partial\theta_1} & \frac{\partial\phi}{\partial\theta_2} \\ \frac{\partial l}{\partial\theta_1} & \frac{\partial l}{\partial\theta_2} \end{bmatrix} = \mathbf{M}_p \frac{\partial \boldsymbol{d}}{\partial\boldsymbol{\theta}} = \mathbf{M}_p\,\mathbf{D}_\theta \tag{11}$$

where $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$, $\mathbf{D}_\theta = \partial\boldsymbol{d}/\partial\boldsymbol{\theta}$, whose off-diagonal elements yield $\frac{\partial d_1}{\partial\theta_2} = \frac{\partial d_2}{\partial\theta_1} = 0$.

The diagonal elements of $\mathbf{D}_\theta$ can be derived by differentiating d_i with respect to θ_i as

$$\frac{\partial d_i}{\partial\theta_i} = \frac{l_2}{1+\cos\theta_i}, \quad i = 1,2 \tag{12}$$

where d_i can be derived as in Eq. (3).

By substituting Eqs. (10) and (11) into (9), the system Jacobian matrix of the studied 2R1T RCM mechanism can be obtained in an analytical form. Note that all the non-zero elements of $\mathbf{J}$ can be derived explicitly and represented in concise forms. Therefore, Jacobian-based analysis, e.g. singularity analysis and performance optimization, can be conducted readily on the studied mechanism.

4 Singularity and Workspace Analysis

In this section, singularity analysis is first conducted on the proposed RCM mechanism, based on the closed-form representation of its Jacobian matrix. Hence, all the singular configurations can be characterized explicitly according to those conditions of the composing One-DOF linkages. As well, the workspace characterization is conducted on the studied RCM mechanism using the analytical models of kinematics. As a result, suitable kinematic parameters can be assigned to the mechanism to achieve a required dexterous workspace.

4.1 Singularity Analysis

In general, a mechanism suffers from singularity when its Jacobian matrix is rank-degenerate. In the studied 2R1T RCM case, this condition yields $\text{rank}(\mathbf{J}) < 3$. Moreover, the system Jacobian matrix is block diagonal since the motion of the DT linkage and the straight-line linkages are decoupled. Then, the condition of singularity could be represented in a concise form as

$$\frac{\partial\theta}{\partial\theta_0} = 0 \quad \text{or} \quad \frac{\partial d_i}{\partial\theta_i} = 0 \tag{13}$$

where the expressions are given in Eqs. (10) and (12).

It is worth noting that, to make the differential motion of the dual-helical joint feasible, the two screw pitches should be different from each other, as $p_1 \neq p_2$. Thus, the transmission matrix $\mathbf{M}_p$ is non-singular, and the lower-right block is singular only when $\det(\mathbf{D}_\theta) = 0$ as Eq. (11). Moreover, the two conditions in Eq. (13) correspond to the singularity of the DT linkage and the straight-line linkages, respectively.

From Eq. (10), the singular configuration relating to the first condition in (13) can be identified directly as

$$\frac{\partial\theta}{\partial\theta_0} = 0 \;\Rightarrow\; \theta_0 + \frac{1}{2}\alpha = \frac{1}{2}\pi \tag{14}$$

which relates to the linkage's limitation of motion, as shown in Fig. 8. It is worth noting that, when $\theta_0 = \pi$, the DT linkage

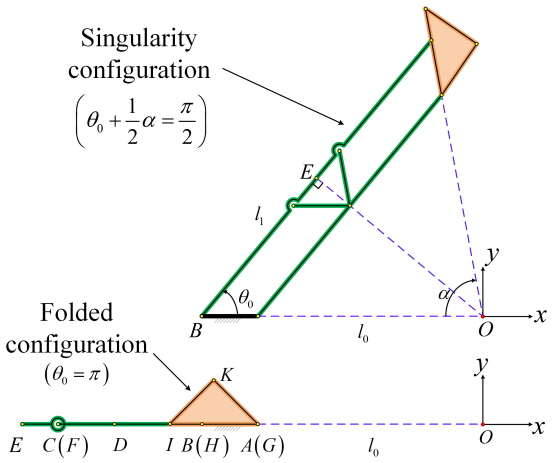


Fig. 8. Singularities of the proposed RCM manipulator

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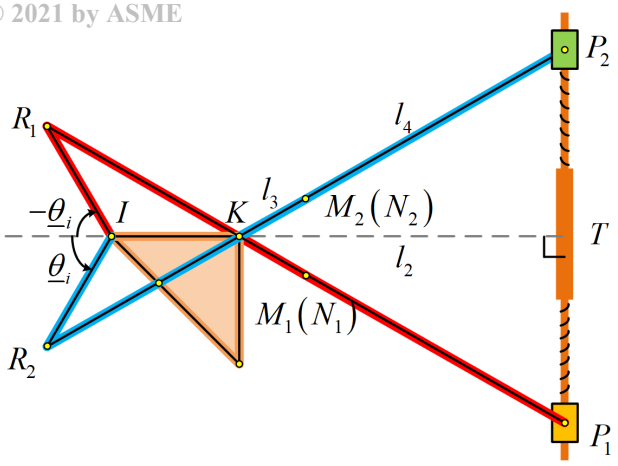


Fig. 9. Singularities of the proposed RCM manipulator

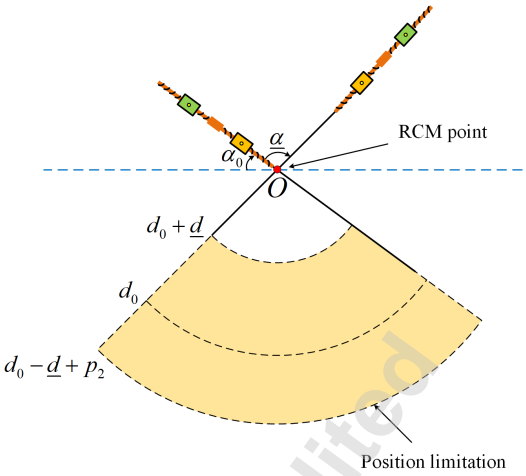


Fig. 10. Reachable worksapce of the proposed mechanism

reaches its dead-point position. In such a case, the whole mechanism folds to a line and is not controllable, either. But, this will not occur in the studied RCM mechanism due to the mechanical interference of the links.

On the other hand, according to Eq. (12), the matrix $\mathbf{D}_\theta$ is full rank within the whose range $\theta_i \in (-\pi, \pi)$. However, the input link IR_i is a rocker that can not perform full-cycle rotation. Therefore, the motion of the straight-line linkage is limited to the range of its dead-point positions, as shown in Fig. 9, which can be determined readily as

$$\underline{\theta}_i = \arccos\left(1 - \frac{(l_4 - l_3)^2}{2l_2^2}\right) \tag{15}$$

where l_3 and l_4 are the lengths of the crank KM_i and the coupler M_iP_i, respectively.

Since the straight-line linkages are symmetric about the base link IK, the valid range of motion of the input links IR_i is within $(-\underline{\theta}_i, \underline{\theta}_i)$. Moreover, the coaxial dual-helical joint will not induce any singularity to the mechanism, because of its constant transmission ratio to the differential motion, as presented in Eq. (4).

From the above analysis, it is obvious that singularity occurs to the proposed RCM mechanism, only when the composing DT or SL linkages are in singularity. For these planar One-DOF linkages, the singular configurations, or more precisely their dead-point positions, can be identified conveniently in geometric methods, as shown in Fig. 8 and 9. And these unexpected positions can be avoided via dimensional synthesis for prescribed ranges of motion.

4.2 Workspace Characterization

As indicated in Sec. 4.1, the composing linkages of the studied RCM mechanism are constrained within the regions between the dead-point positions, in which the mechanism suffers from singularity. Thus, the valid ranges of motion of these One-DOF linkages can then be determined readily, in an individual manner.

First, according to Eq. (5), the rolling angle of the mechanism's end-effector will be confined within the range of

$$\theta \in [\alpha_0,\ \alpha_0 + \underline{\alpha}] \tag{16}$$

where $\underline{\alpha} = 2\arctan\frac{l_1}{l_0}$ corresponds to the dead-point position shown in Fig. 8.

Obviously, the end-effector of the RCM mechanism is of the rolling stroke $\underline{\alpha}$ which is a function exclusively of the ratio between l_1 and l_0. On the other hand, the inclined angle α_0 introduces an offset to the range of end-effector rolling, as shown in Fig. 10. Consequently, this range of motion can be adjusted by choosing the corresponding parameters in an appropriate way.

To ease the implementation of the RCM mechanism, one of the pitches in the dual-helical joint is set to zero as $p_1 = 0$. Since the end-effector's spinning and insertion are coupled, a difference between d_1 and d_2, namely $\Delta d = d_2 - d_1 = p_2$, is set aside to achieve a full-cycle spinning. Hence, the valid stroke of insertion decreases to $2\underline{d} - p_2$, in the range of

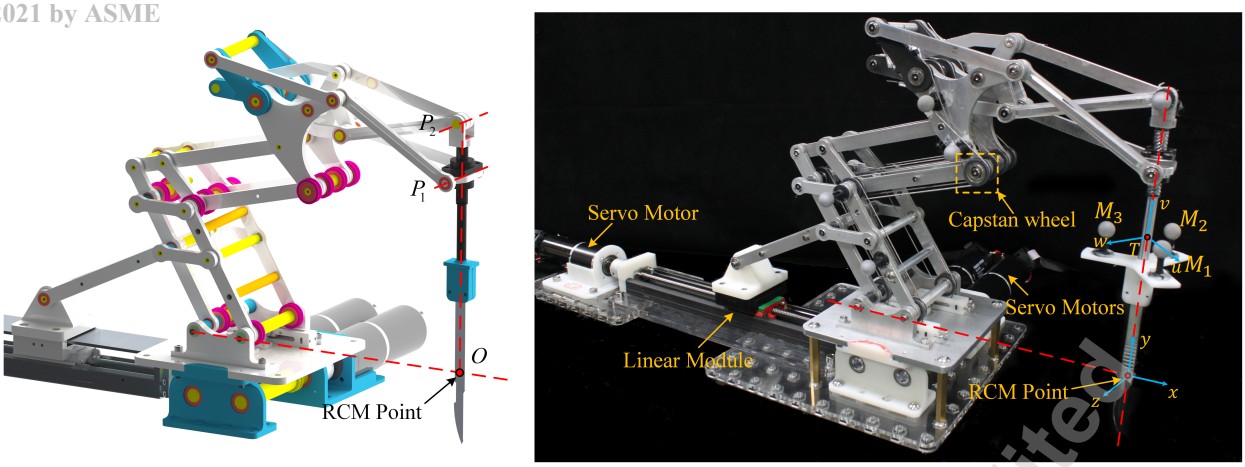
$$l \in [d_0 - \underline{d} + p_2,\ d_0 + \underline{d}] \tag{17}$$

where $\underline{d}$ is the motion limitation of the straight line linkages along the end-effector. It can be determined according to the dead-point positions, given in Eq. (15) as $\underline{d} = l_2 \tan\frac{\theta_i}{2}$.

Similar to the rolling of the end-effector, the range of insertion depends on the initial position d_0 and the stroke $2\underline{d} - p_2$, as illustrated in Fig. 10. The initial position d_0 can be changed conveniently by means of adjusting the tool frame along the end-effector. On the other hand, the motion stroke is a function of the kinematic parameters of the straight-line linkage and the pitches of the dual-helical joint, including $\{l_2, l_3, l_4\}$ and $\{p_1, p_2\}$. Hence, in actual applications, these parameters need to be synthesized for the RCM mechanism to perform the prescribed tasks. In this paper, the prototype is developed to verify the feasibility of the proposed design for RCM realization. Hence, those kinematic parameters are not designed strictly for actual applications.

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(a) (b)

Fig. 11. A prototype of the proposed RCM mechanism.

5 Prototyping and Experimental Validation

To verify the feasibility of the proposed design, a prototype of the proposed 2R1T RCM mechanism is developed in this section, as shown in Fig. 11. Preliminary experiments are conducted on the prototype to characterize the reachable workspace and positioning accuracy. The results exhibit the potential of the particular design for RCM mechanisms.

Table 1. kinematic parameters of the developed prototype.

Para	l_0	l_1	l_2	l_3
value	225 *mm*	143.75 *mm*	127.28 *mm*	30 *mm*
Para	l_4	r	α_0	
value	116.74 *mm*	1.25	45°	

5.1 Prototyping and Experimental Setup

As illustrated in Fig. 11, a scale-down prototype of the proposed RCM mechanism is designed in detail, and then developed using easy-to-access materials and simple fabrication methods. As shown in the figure, coupling-supported structures are adopted to enhance the structural stiffness of the composing links, keeping low inertia. 3D-printed plastic parts are fabricated for the components with complex shapes. Besides, commercial ball screw-and-nut is employed for the dual-helical joint to generate the differential motion for end-effector spinning and insertion.

To simplify the actuation of the composing linkages, a planar slider-crank mechanism is introduced to control the input variable θ_0 of the dual-triangular linkage, as shown in Fig. 11. Meanwhile, a linear module, driven by a high-quality servo motor (Maxon RE40), is adopted to actuate the slider-crank mechanism precisely. In addition, cable-driven technique is employed to actuate the input links of the straight-line linkages. Here, Capstan wheels are used to transmit the rotation of the driving motors to the straight-line linkages through the compact cable-driven systems. As a result, all the actuation motors can be mounted on the base frame of the mechanism and the inertia of the linkages' moving links can be kept in a low level.

In the experiential apparatus, the motion capture system 'OptiTrack' is integrated to measure the end-effector pose of the RCM mechanism. As shown in Fig. 11, three spherically mounted reflectors (SMRs), labeled as M_1, M_2 and M_3, are attached to the end-effector to construct the tool frame $\{\mathrm{T}\}$. Note that these three SMRs are arranged in an equilateral triangle on a plane perpendicular to the distal axis. Thus, the tool frame can be set up in a convenient way, by locating the origin at the center of SMRs, and orienting its v-axis normal to the SMRs plane, and letting the u-axis pass through M_1. As a result, the 2R1T motion of the developed RCM prototype can be characterized by the rotations about the v-axis and u-axis of $\{\mathrm{T}\}$, and the translation along the v-axis. These data can be collected conveniently by capturing the SMRs using the OptiTrack system.

As indicated in Sec. 4.2, the reachable workspace of the studied RCM mechanism depends on several key parameters. In the developed prototype, these kinematic parameters are determined in light of a prescribed workspace. As shown in Fig. 10, the rolling angle of the mechanism's end-effector is set to $\theta \in [70°, 120°]$, by referring to the literature [28, 29]. Meanwhile, the screw pitches of the dual-helical joint are set to $p_1 = 0\,mm$ and $p_2 = 20\,mm$, respectively, and the required insertion of the end-effector is set to $\underline{d} \in [200mm, 250mm]$. It is worth noting that $\underline{d}$ relates to the distance from P_1 to T, as $p_1 = 0$, which is slightly different from the one in Eq. (17). Since the motion range of PL linkages covers over the pitch difference, the end-effector can perform full-cycle spinning.

According to the workspace characterization presented in Sec. 4.2, the kinematic parameters corresponding to the above required workspace can be identified for the prototype, as listed in Table. 1. It should be noted that, the scale-down prototype is not particularly developed for a specific application, but a preliminary verification of the design proposed in this paper. Hence, the reachable workspace of the prototype may probably be smaller than that required for actual MIS.

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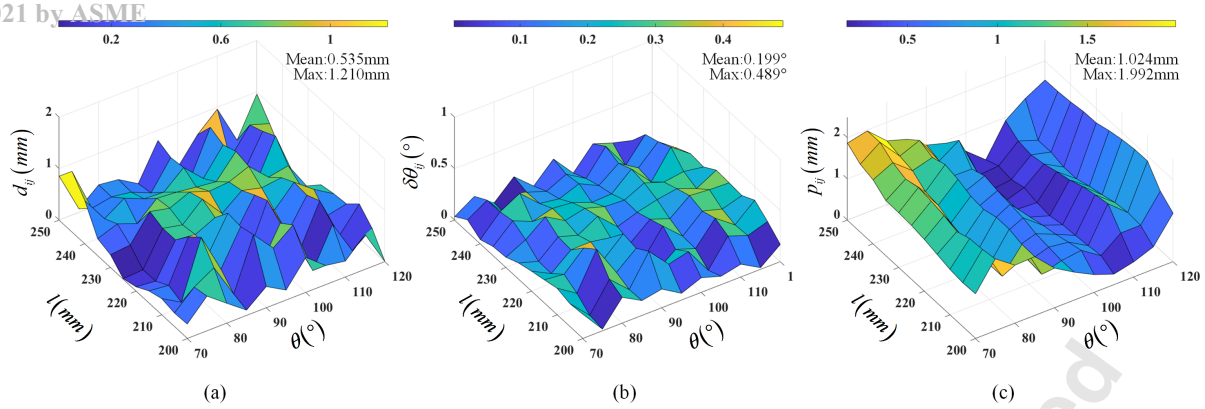


Fig. 12. Accuracy evaluation of the developed prototype: (a) misalignment of the end-effector to the mechanism's RCM point, (b) angular deviation of the end-effector's actual orientations from nominal, (c) translational inaccuracy along the axial direction

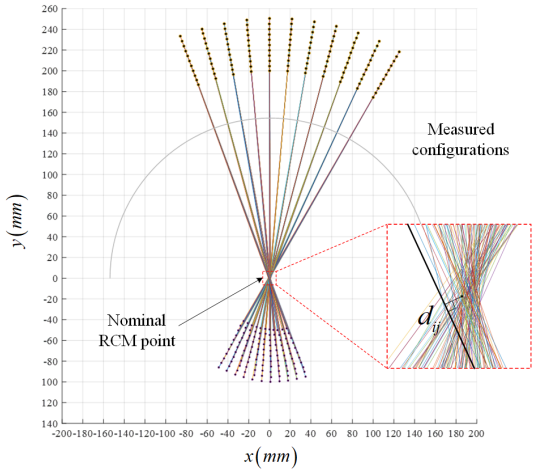


Fig. 13. Measured configurations of the end-effector.

5.2 Experimental Validation

In this section, validation experiments are conducted on the developed prototype of the studied RCM mechanism. The workspace that the mechanism's end-effector can reach is characterized by discretizing the rolling and insertion. As well, the positioning accuracy of the end-effector is assessed to verify the effectiveness of the proposed design and potential applications in actual MIS.

As indicated in Sec. 5.1, the OptiTrack system is utilized in the experimental apparatus to measure the end-effector pose of the developed prototype. Since the 2R1T motion of the proposed RCM mechanism is partially decoupled, the validation of workspace is conducted by means of discretizing the range of rolling, $[70°, 120°]$, into 11 orientations with the increment of $5°$. For each orientation, the range of axial insertion, $[200mm, 250mm]$, is also discretized into 11 positions with the increment of $5mm$. As a result, $11 \times 11 = 121$ configurations are totally generated for the end-effector in this experiment. According to the kinematics model in Sec. 3.2, the corresponding inputs can be derived readily for the actuators. A protocol between the motor controller and the OptiTrack system is built, such that these discretized configurations can be actuated and then measured automatically.

The results are given in Fig. 13. It is worth noting that, in the figure, there are 11 bunches of radial lines, with their tip points marked with circles. The tip point corresponds to the position of the nut of the zero-pitch screw in the dual-helical joint, namely the point P_1 as indicated in Fig. 11. The directions of these radial lines are coincident with the v-axis of the tool frame, which is perpendicular to the setting plane of the measuring SMRs. Theoretically, all these lines should pass through the nominal RCM point strictly. However, due to the manufacturing and assembling errors, there does not exist an exact common point for the end-effector when the mechanism moves. And the divergence of these lines relates to the quality of the end-effector RCM characteristics of the developed prototype.

In order to evaluate the RCM characteristics of the prototype's end-effector, the distance from the end-effector axis to the nominal RCM point is extracted for each measured configuration illustrated in Fig. 13. As shown in the figure, d_{ij} represents the distance for the ij^{th} configuration. And the distribution of all these distances within the measured configurations (discretized by the rolling and insertion motion) is illustrated in Fig. 12(a). From the results, it is known that the mean and maximum values of the axial misalignment are $0.54mm$ and $1.21mm$, respectively, within the prescribed workspace. To some extent, these magnitudes are reasonably acceptable for a preliminary prototype using easy-to-access material and simple fabrication methods.

The absolute positioning accuracy of the end-effector's rolling and insertion motion is also evaluated using the data, collected in the validation experiments. First, the deviations of the measured orientations from the prescribed nominal ones are calculated as the angular errors of rolling. The distribution of this error within the measured configurations are drawn in Fig. 12(b). It can be seen that these angular errors are distributed evenly throughout the measured workspace, with $0.20°$ and $0.49°$ as the mean and the maximum values, respectively. And it is also acceptable considering the simple manufacturing methods used for the prototype.

Besides, the linear errors of the end-effector's insertion along the corresponding central axes are also calculated for the validation of the prototype's positioning accuracy. The

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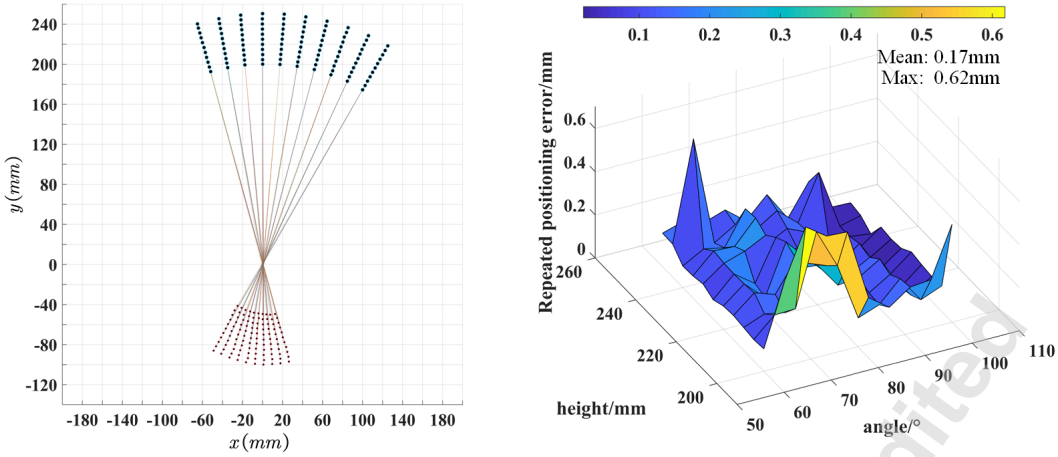


Fig. 14. Repeatability of the developed prototype within the prescribed workspace.

distribution of these translational errors is illustrated in Fig. 12(c). The mean and maximum values of this translational error are 1.02*mm* and 1.99*mm*, respectively. Obviously, this error is a little bit large comparing to the axial misalignment of the end-effector from the RCM point. Here, this is mainly caused by the transmission error of the cable-driven systems, employed for the actuation of the straight-line linkages. Generally, this is comparatively larger than that generated in the rigid-body linkages.

Furthermore, the repeatability analysis of the developed RCM prototype is also conducted in this section. As shown in Fig. 14, the prescribed workspace is discretized about the rolling and insertion motion. The mean and maximum values of repeatability are 0.17*mm* and 0.62*mm*, respectively, over the prototype's workspace. From the results, it is shown that the mean value of repeatability is about one order of magnitude higher than the one of absolute positioning accuracy. It is reasonable because many error sources are not taken into account in the control theme of the prototype. As it is known that, due to unexpected aspects, such as cable tension and pulley friction, the transmission of the cable-driven systems is not as precise as the rigid-body linkages are.

It should be noted, in the validation experiments, a couple of extreme configurations can not be reached due to the mechanical interference of the composing linkages and the cable-driven systems. Further, due to the manufacturing and assembling errors, the positioning accuracy of the developed prototype is not as high as required in the actual MIS, either. As addressed in the above section, the prototype developed in this paper serves as a preliminary experimental apparatus to verify the feasibility of the proposed idea for designing RCM mechanisms. In our future work, better prototypes will be particularly developed according to the requirements of specific MIS application. High-performance materials and precise fabrication techniques can be adopted to improve the performance of the upgraded RCM mechanisms. Moreover, kinematic calibration can then be conducted to improve the positioning accuracy, by means of compensating the manufacturing and assembling errors in the control theme.

6 Conclusion

In this paper, a novel type of planar RCM mechanism with the 2R1T mobility is proposed by composing specific one-DOF closed-loop linkages in a consecutive way. First, a dual-triangular linkage is utilized to achieve the rolling of the mechanism's end-effector about a remote stationary point. Meanwhile, two Peaucellier-Lipkin straight-line linkages are employed to realize the coaxial spinning and insertion of the end-effector, through the differential motion of a dual-helical joint. Thanks to the simplicity in structure, kinematic models of the proposed mechanism can be derived in closed form, with simpler reachable workspace and singular configurations. A preliminary prototype is developed utilizing easy-to-access materials and simple fabrication methods, on which various validation experiments are conducted. The results show that, the developed prototype is easy to control and capable of performing 2R1T RCM manipulation with acceptable positioning accuracy. Moreover, the proposed design of planar 2R1T RCM mechanism can be extended to a spatial 3R1T version conveniently. Therefore, the proposed RCM mechanism has the potential of application in robot-assisted minimally invasive surgeries.

Acknowledgements

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